

DATA STRUCTURES

LAB PROGRAMS

23.Prim's algorithm

```
#include <stdio.h>
```

```
#define V 4
```

```
int main() {
```

```
    int graph[V][V] = {
```

```
        {0,2,0,6},
```

```
        {2,0,3,8},
```

```
        {0,3,0,0},
```

```
        {6,8,0,0}
```

```
    };
```

```
    int selected[V]={0};
```

```
    selected[0]=1;
```

```
    int edge=0;
```

```
    printf("Edges in MST:\n");
```

```
    while(edge < V-1) {
```

```
        int min=999,a=0,b=0;
```

```
        for(int i=0;i<V;i++) {
```

```
            if(selected[i]) {
```

```
                for(int j=0;j<V;j++) {
```

```
                    if(!selected[j] && graph[i][j]) {
```

```
                        if(min > graph[i][j]) {
```

```

        min = graph[i][j];
        a=i;
        b=j;
    }
}
}
}
}
printf("%d - %d : %d\n",a,b,min);
selected[b]=1;
edge++;
}
return 0;
}

```

Output

Edges in MST:

0 - 1 : 2

1 - 2 : 3

0 - 3 : 6

=== Code Execution Successful ===